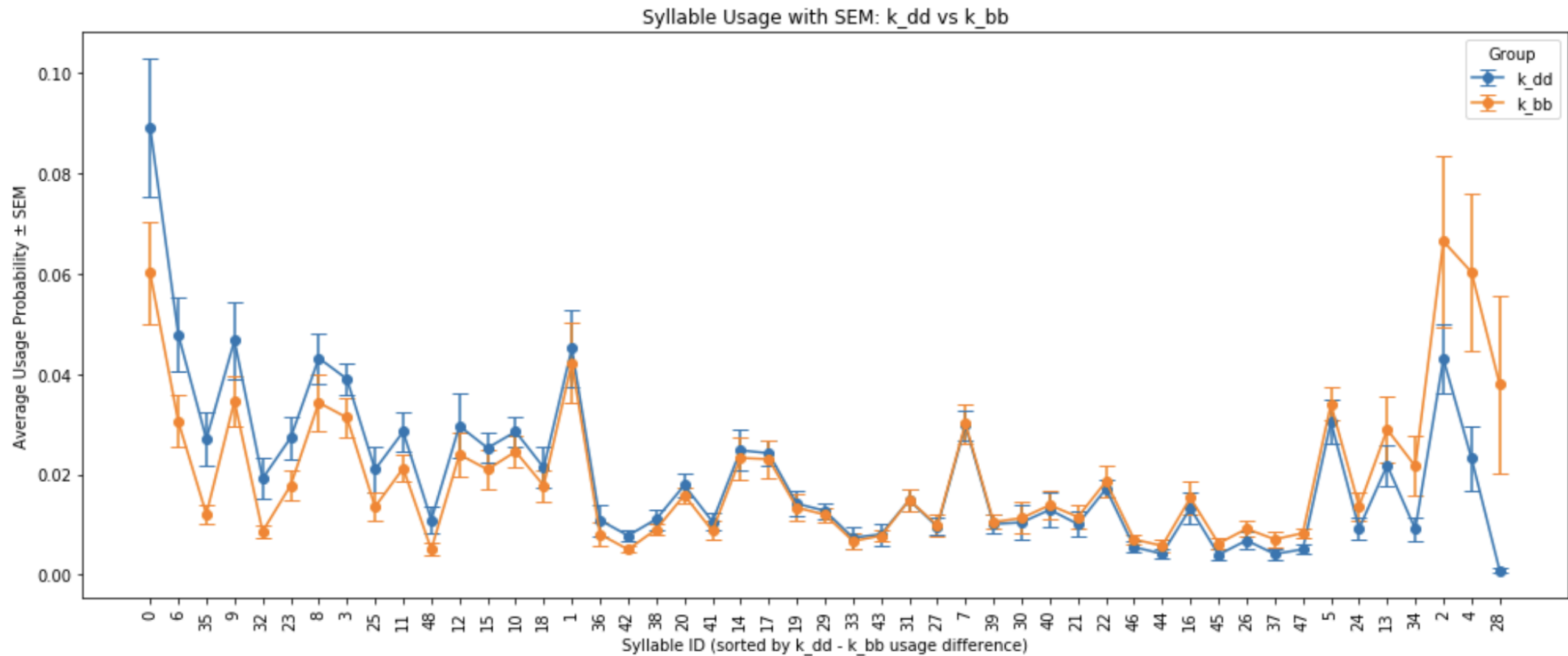


07/11 Update

Syllable Usage

2 Hour after injection (k_dd) vs Baseline (k_bb)



Syllable t-test with FDR adjusted p-value

	syllable	t_stat	p_value	fdr_p_value	significant
35	35	2.630517	0.016405	0.362212	False
32	32	2.508539	0.022042	0.362212	False
47	47	-2.283081	0.028128	0.362212	False
4	4	-2.176910	0.037489	0.362212	False
28	28	-2.090028	0.047855	0.362212	False
42	42	2.060128	0.049929	0.362212	False
34	34	-1.969547	0.058309	0.362212	False
6	6	1.953778	0.060369	0.362212	False
23	23	1.803332	0.081472	0.434515	False
0	0	1.698257	0.099787	0.478978	False
3	3	1.555827	0.128051	0.491343	False
37	37	-1.524026	0.135809	0.491343	False
45	45	-1.521082	0.137573	0.491343	False
11	11	1.504880	0.143309	0.491343	False
25	25	1.413320	0.168947	0.540632	False
9	9	1.318867	0.198410	0.570226	False
2	2	-1.276504	0.211555	0.570226	False
24	24	-1.264268	0.213835	0.570226	False
8	8	1.168926	0.249779	0.631021	False
46	46	-1.011151	0.318673	0.725630	False
26	26	-0.988494	0.329479	0.725630	False
44	44	-0.955444	0.345395	0.725630	False
13	13	-0.927512	0.359849	0.725630	False
10	10	0.921222	0.362815	0.725630	False
15	15	0.879825	0.384495	0.738230	False
38	38	0.847721	0.404518	0.746802	False
12	12	0.740050	0.465426	0.763134	False

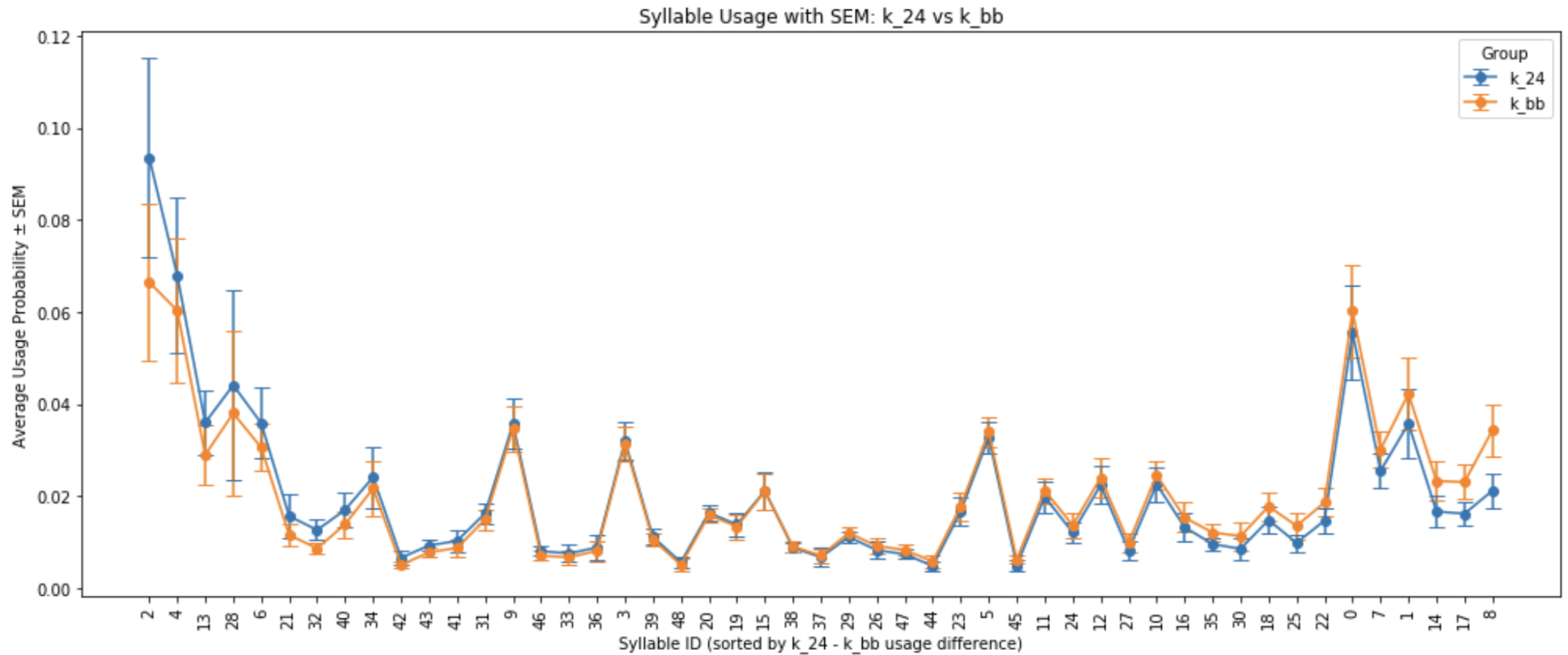
Syllable test between k_dd group and k_bb group, which is the ketamine injection group after 2 hours compared with the baseline group.

The p-value for syllable 35, 32, 47, 4, 28 42 is less than 0.05. However, the FDR p-values are all non-significant.

I used Benjamini-Hochberg's method for the FDR

Syllable Usage

24 Hour after injection (k₂₄) vs Baseline (k_{bb})



Syllable t-test with FDR adjusted p-value

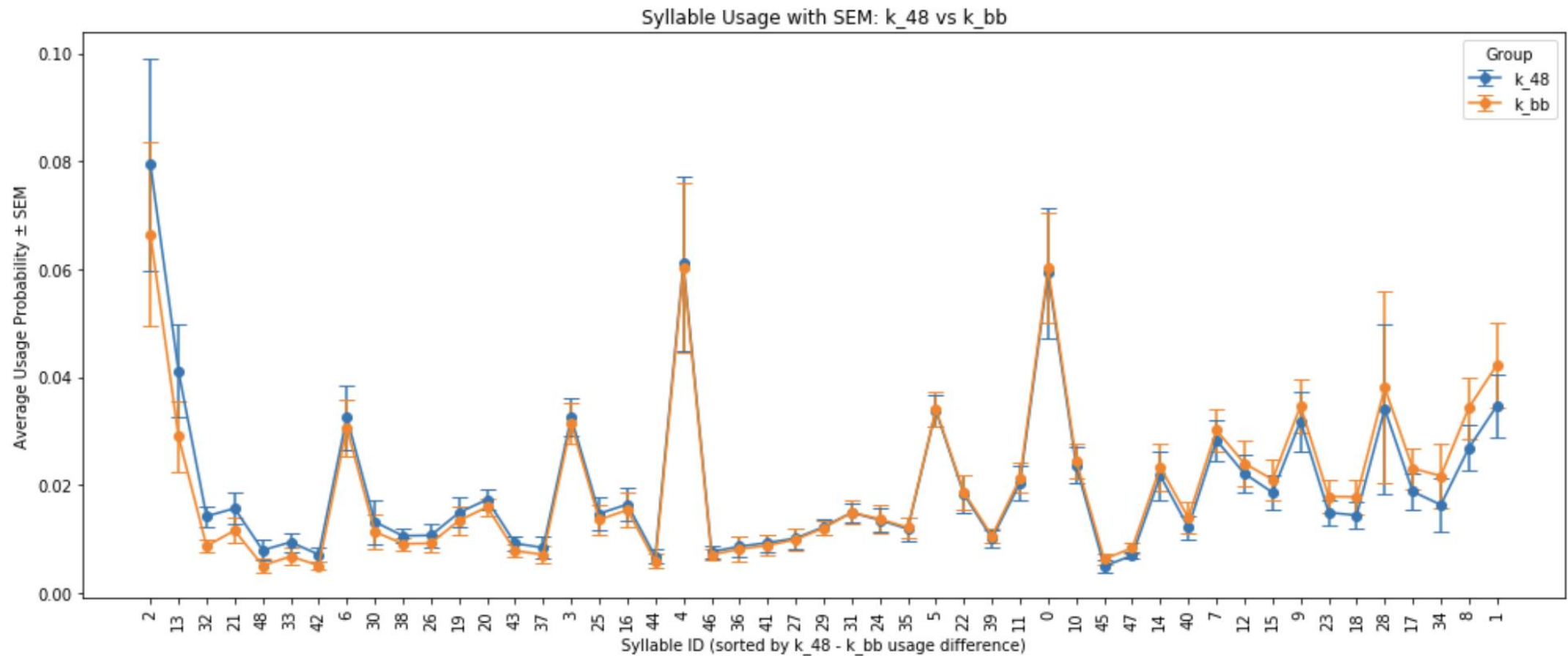
	syllable	t_stat	p_value	fdr_p_value	significant
8	8	-1.920845	0.061897	0.933040	False
32	32	1.600728	0.118502	0.933040	False
17	17	-1.491458	0.143453	0.933040	False
14	14	-1.195157	0.238462	0.933040	False
25	25	-1.076436	0.288260	0.933040	False
35	35	-1.053056	0.298662	0.933040	False
2	2	0.980858	0.332271	0.933040	False
42	42	0.978598	0.335282	0.933040	False
22	22	-0.959023	0.342734	0.933040	False
43	43	0.925703	0.359595	0.933040	False
45	45	-0.891957	0.377284	0.933040	False
7	7	-0.855035	0.397061	0.933040	False
21	21	0.793128	0.433535	0.933040	False
13	13	0.732394	0.467749	0.933040	False
30	30	-0.699921	0.487796	0.933040	False
18	18	-0.684717	0.497037	0.933040	False
40	40	0.684453	0.497448	0.933040	False
46	46	0.682821	0.498220	0.933040	False
47	47	-0.617842	0.539795	0.933040	False
1	1	-0.590402	0.557878	0.933040	False
44	44	-0.586992	0.560337	0.933040	False
27	27	-0.573245	0.569351	0.933040	False
6	6	0.570256	0.571807	0.933040	False
41	41	0.483853	0.631014	0.933040	False
29	29	-0.475661	0.636630	0.933040	False
16	16	-0.470404	0.640339	0.933040	False
24	24	-0.453987	0.652134	0.933040	False
31	31	0.427253	0.671233	0.933040	False

Syllable test between k_24 group and k_bb group, which is the ketamine injection group after 24 hours compared with the baseline group.

None of the syllable has significant usage difference.

Syllable Usage

48 Hour after injection (k_48) vs Baseline (k_bb)



Syllable t-test with FDR adjusted p-value

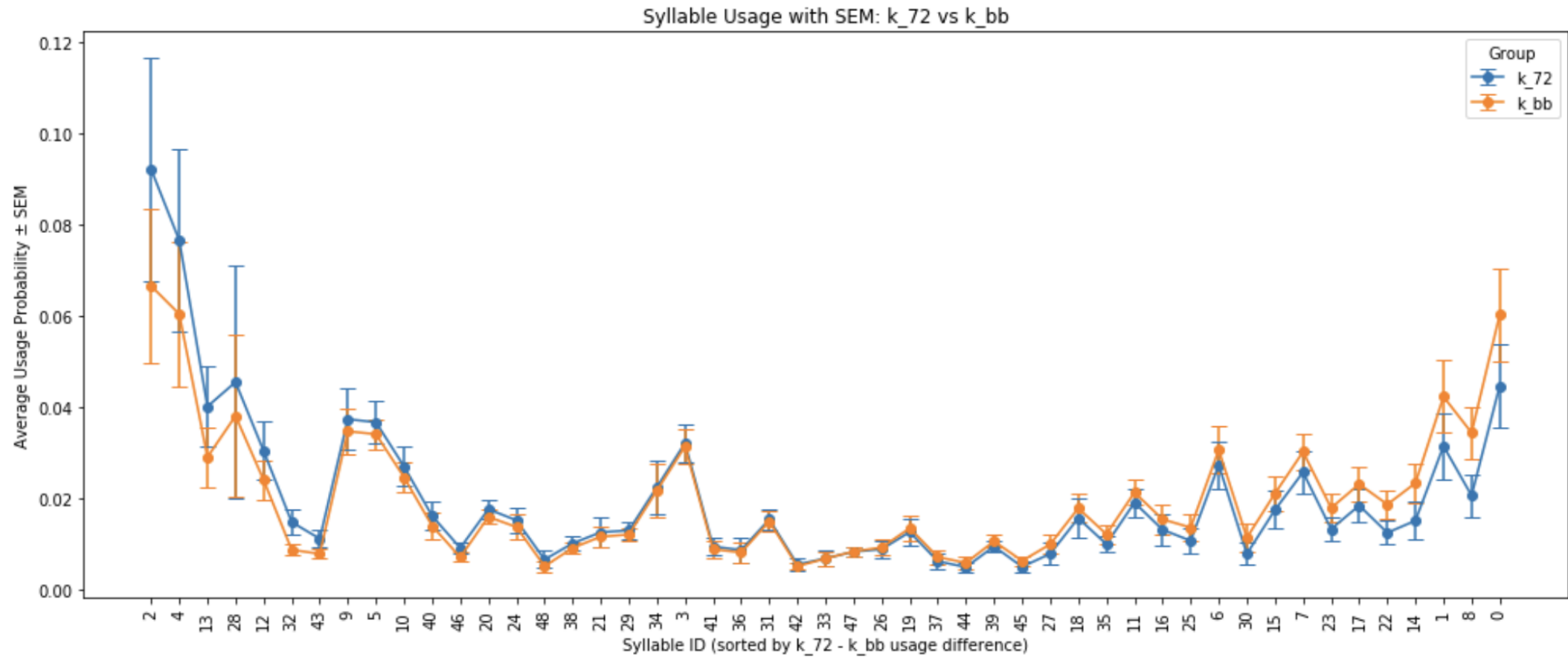
	syllable	t_stat	p_value	fdr_p_value	significant
32	32	2.485200	0.017272	0.829075	False
42	42	1.303060	0.201579	0.991359	False
13	13	1.122752	0.267779	0.991359	False
21	21	1.117385	0.269932	0.991359	False
33	33	1.111214	0.272308	0.991359	False
47	47	-1.082398	0.285635	0.991359	False
8	8	-1.052064	0.298682	0.991359	False
45	45	-0.932265	0.356098	0.991359	False
17	17	-0.858767	0.394990	0.991359	False
18	18	-0.855071	0.397227	0.991359	False
38	38	0.814940	0.419350	0.991359	False
43	43	0.775275	0.442371	0.991359	False
1	1	-0.765374	0.448306	0.991359	False
23	23	-0.760728	0.450977	0.991359	False
34	34	-0.696095	0.489998	0.991359	False
20	20	0.578940	0.565539	0.991359	False
26	26	0.551152	0.584357	0.991359	False
37	37	0.507915	0.614107	0.991359	False
2	2	0.499178	0.620082	0.991359	False
40	40	-0.495209	0.622965	0.991359	False
44	44	0.480415	0.633211	0.991359	False
15	15	-0.472207	0.639055	0.991359	False
9	9	-0.378874	0.706550	0.991359	False
19	19	0.376807	0.708050	0.991359	False
46	46	0.357991	0.721990	0.991359	False
30	30	0.355739	0.723771	0.991359	False
7	7	-0.344164	0.732295	0.991359	False
12	12	-0.334457	0.739606	0.991359	False
35	35	0.270211	0.780110	0.991359	False

Syllable test between k_48 group and k_bb group, which is the ketamine injection group after 48 hours compared with the baseline group.

Syllable 32 has a p-value less than 0.05, but FDR p-value is non-significant.

Syllable Usage

72 Hour after injection (k_72) vs Baseline (k_bb)



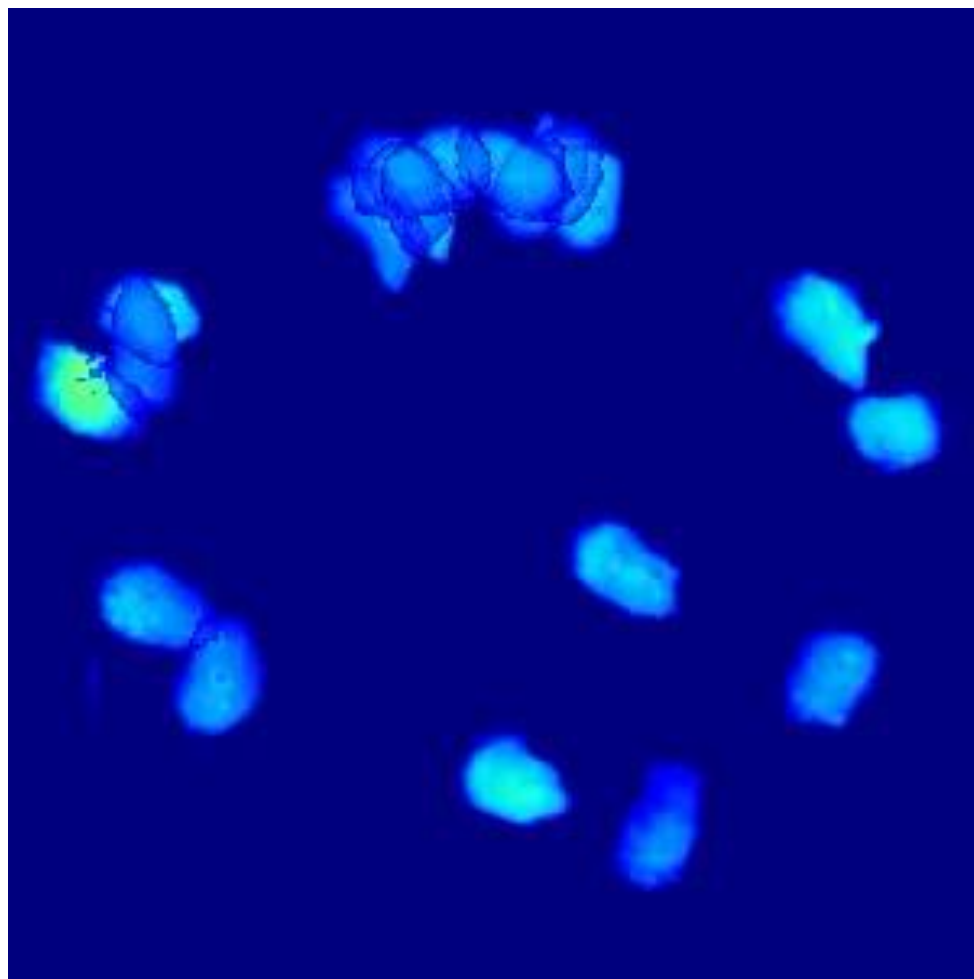
Syllable t-test with FDR adjusted p-value

	syllable	t_stat	p_value	fdr_p_value	significant
32	32	2.052088	0.051261	0.956446	False
8	8	-1.886980	0.066460	0.956446	False
22	22	-1.542142	0.130918	0.956446	False
43	43	1.521565	0.139271	0.956446	False
14	14	-1.376875	0.176348	0.956446	False
46	46	1.321390	0.194137	0.956446	False
23	23	-1.161449	0.252372	0.956446	False
0	0	-1.146427	0.258435	0.956446	False
13	13	1.020191	0.314887	0.956446	False
1	1	-1.001645	0.322562	0.956446	False
45	45	-0.900898	0.373421	0.956446	False
17	17	-0.899782	0.373691	0.956446	False
30	30	-0.899788	0.373710	0.956446	False
2	2	0.854397	0.399283	0.956446	False
35	35	-0.849465	0.400680	0.956446	False
12	12	0.835899	0.409513	0.956446	False
7	7	-0.741893	0.462906	0.956446	False
25	25	-0.716496	0.477946	0.956446	False
20	20	0.679071	0.501577	0.956446	False
27	27	-0.650423	0.519377	0.956446	False
15	15	-0.633815	0.530009	0.956446	False
4	4	0.632058	0.531497	0.956446	False
39	39	-0.603509	0.549580	0.956446	False
44	44	-0.587829	0.560054	0.956446	False
11	11	-0.534394	0.596249	0.956446	False
38	38	0.532626	0.597572	0.956446	False
40	40	0.518214	0.607342	0.956446	False
16	16	-0.488008	0.628311	0.956446	False

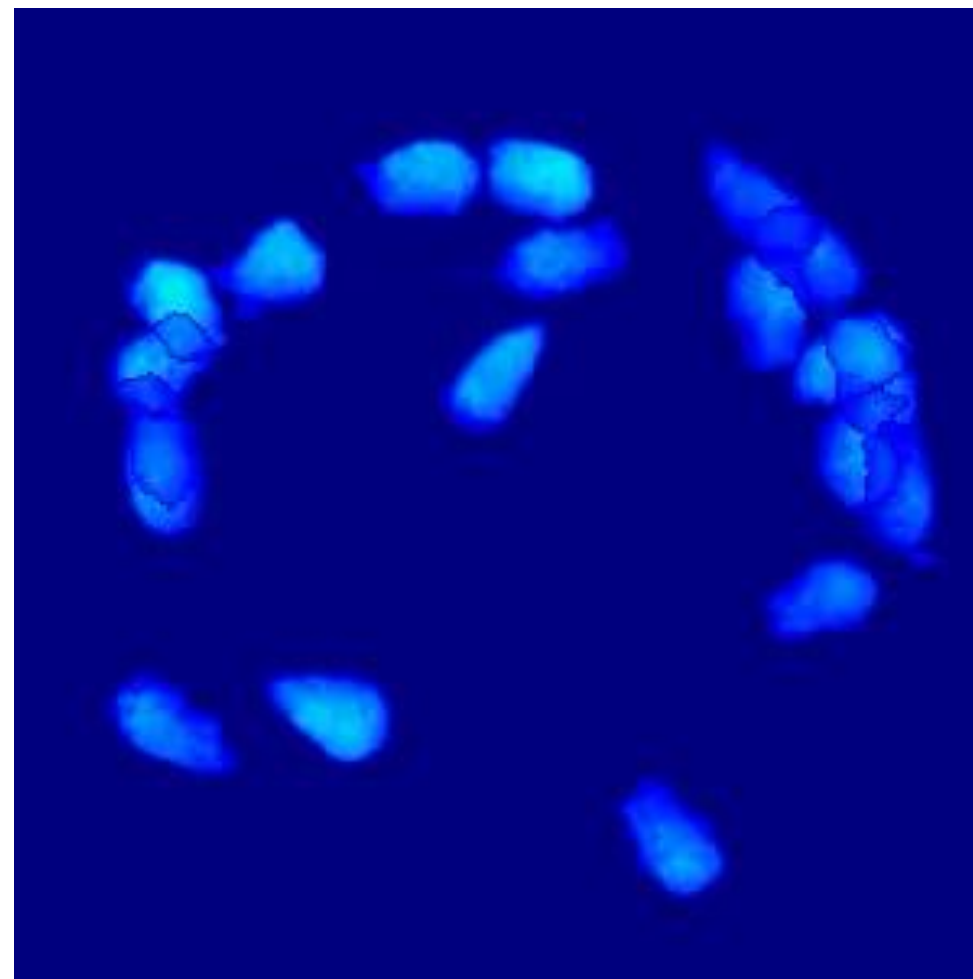
Syllable test between k_72 group and k_bb group, which is the ketamine injection group after 72 hours compared with the baseline group.

None of the syllable has significant usage difference.

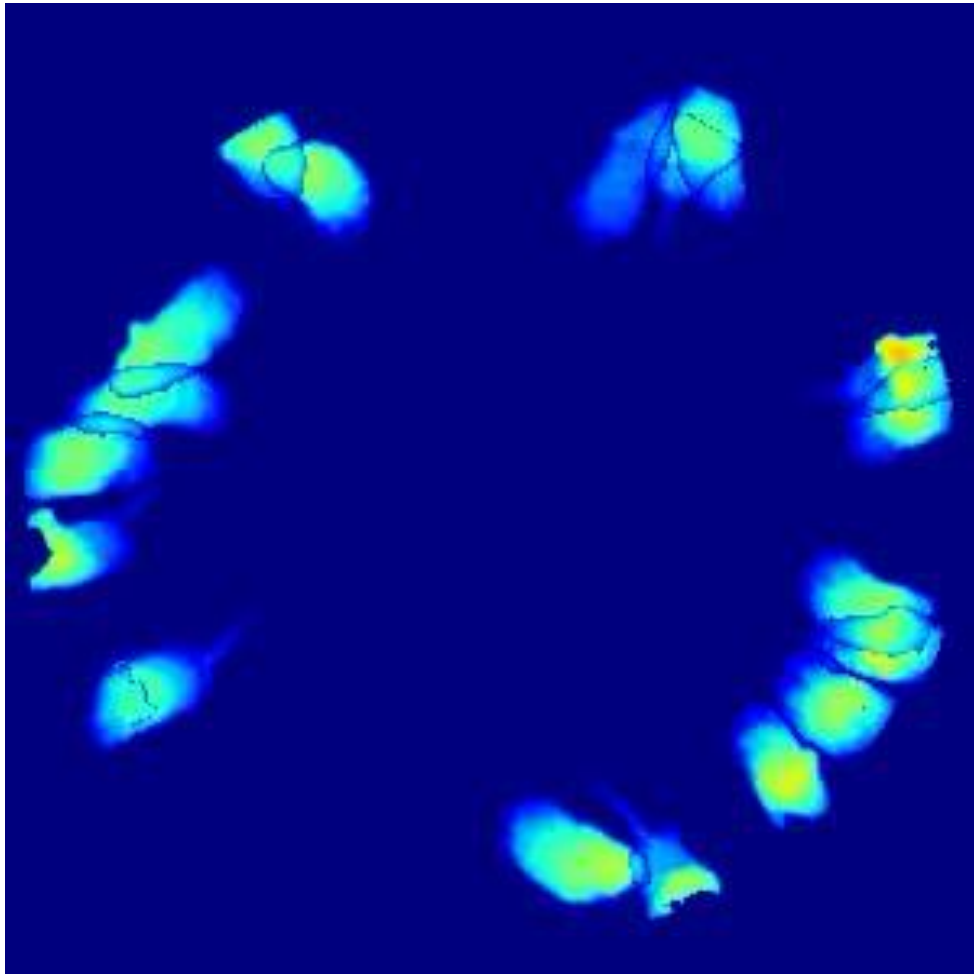
Syllable 32



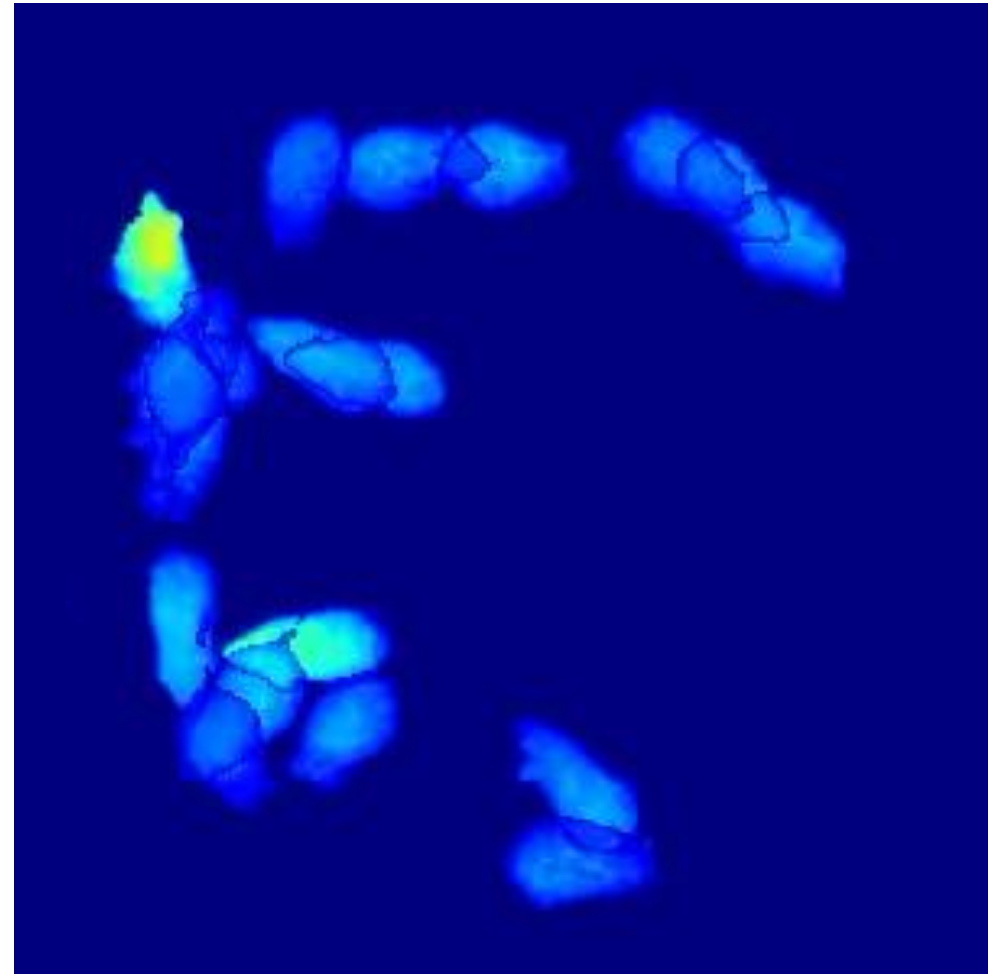
Syllable 35



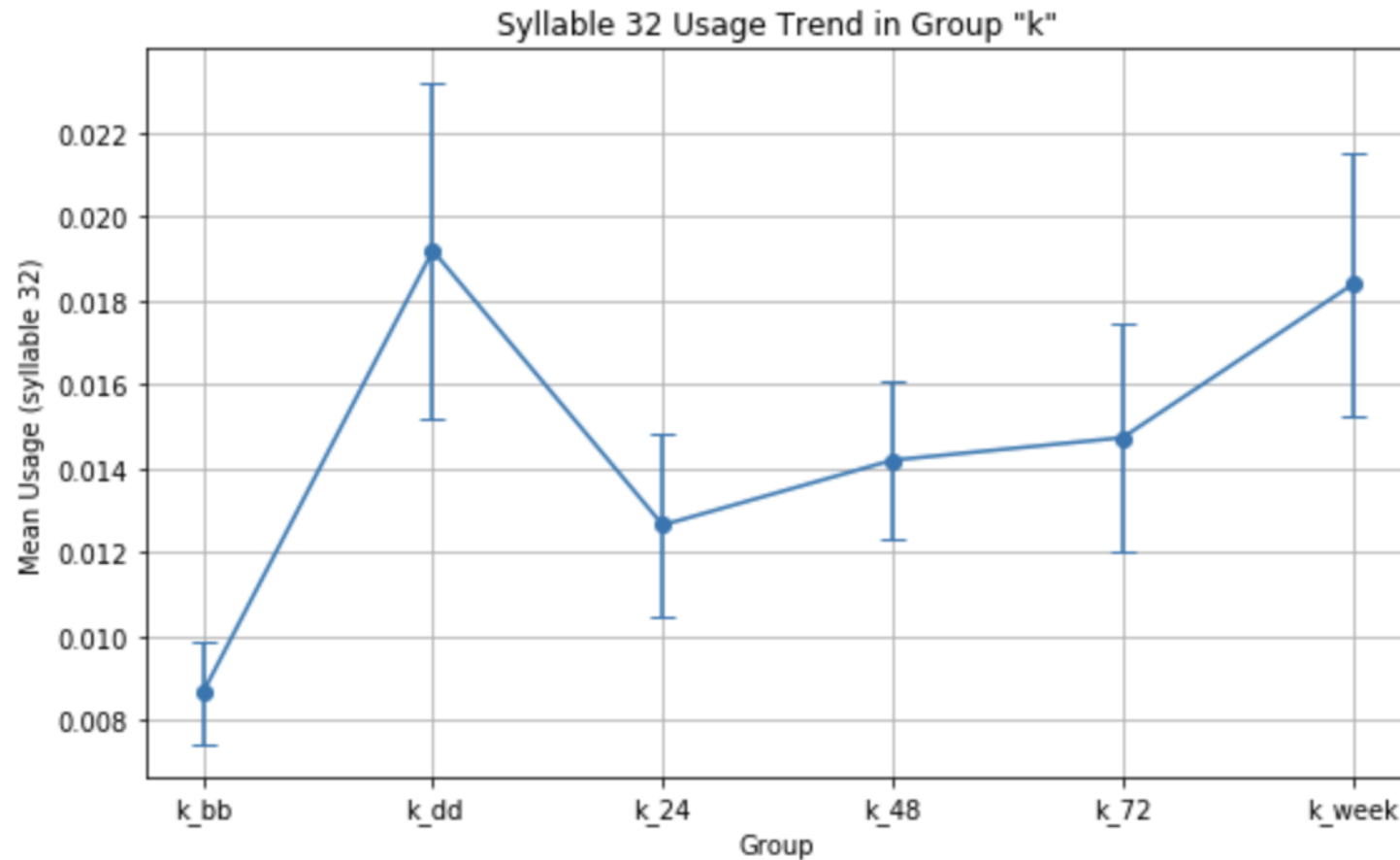
Syllable 47



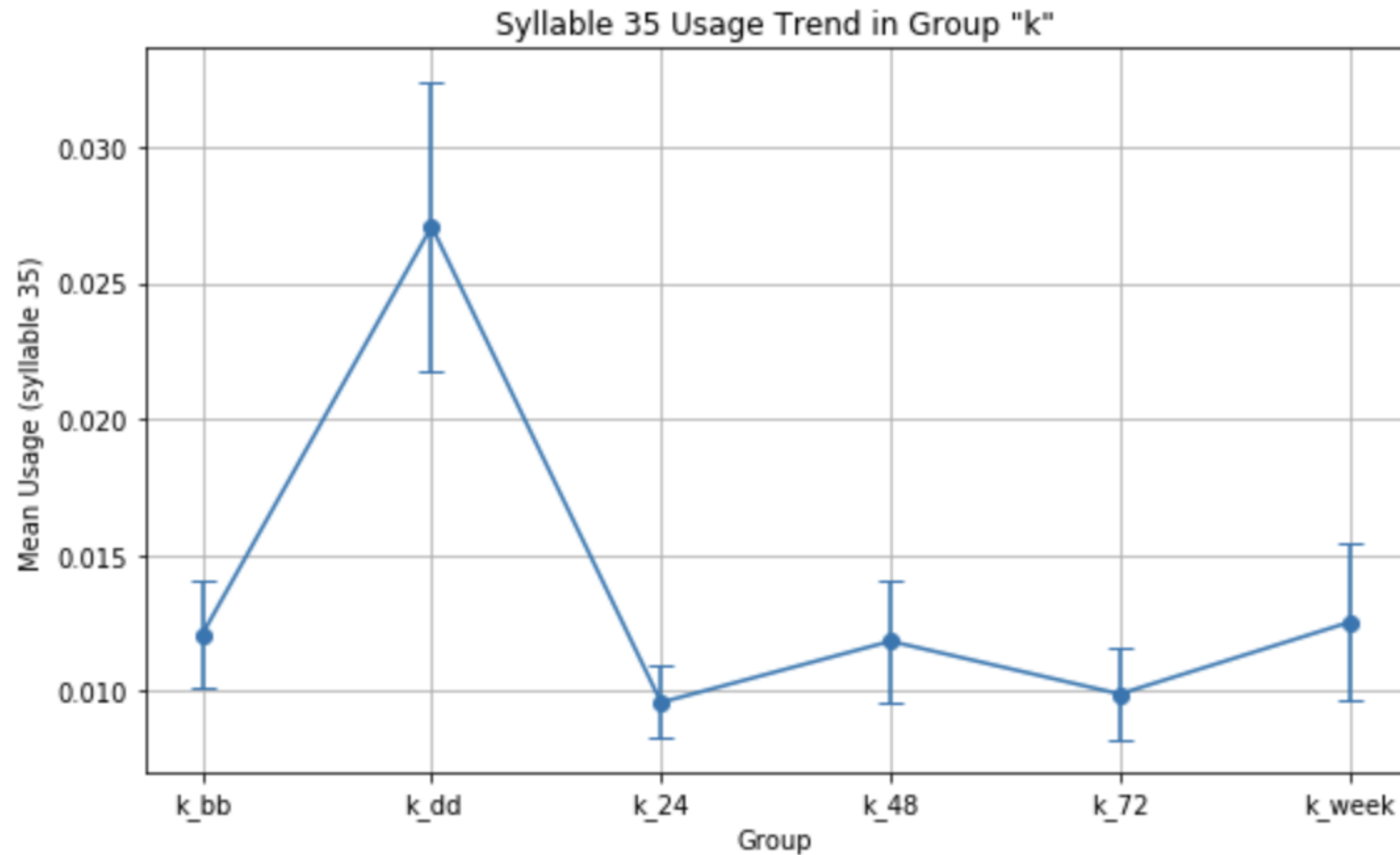
Syllable 4



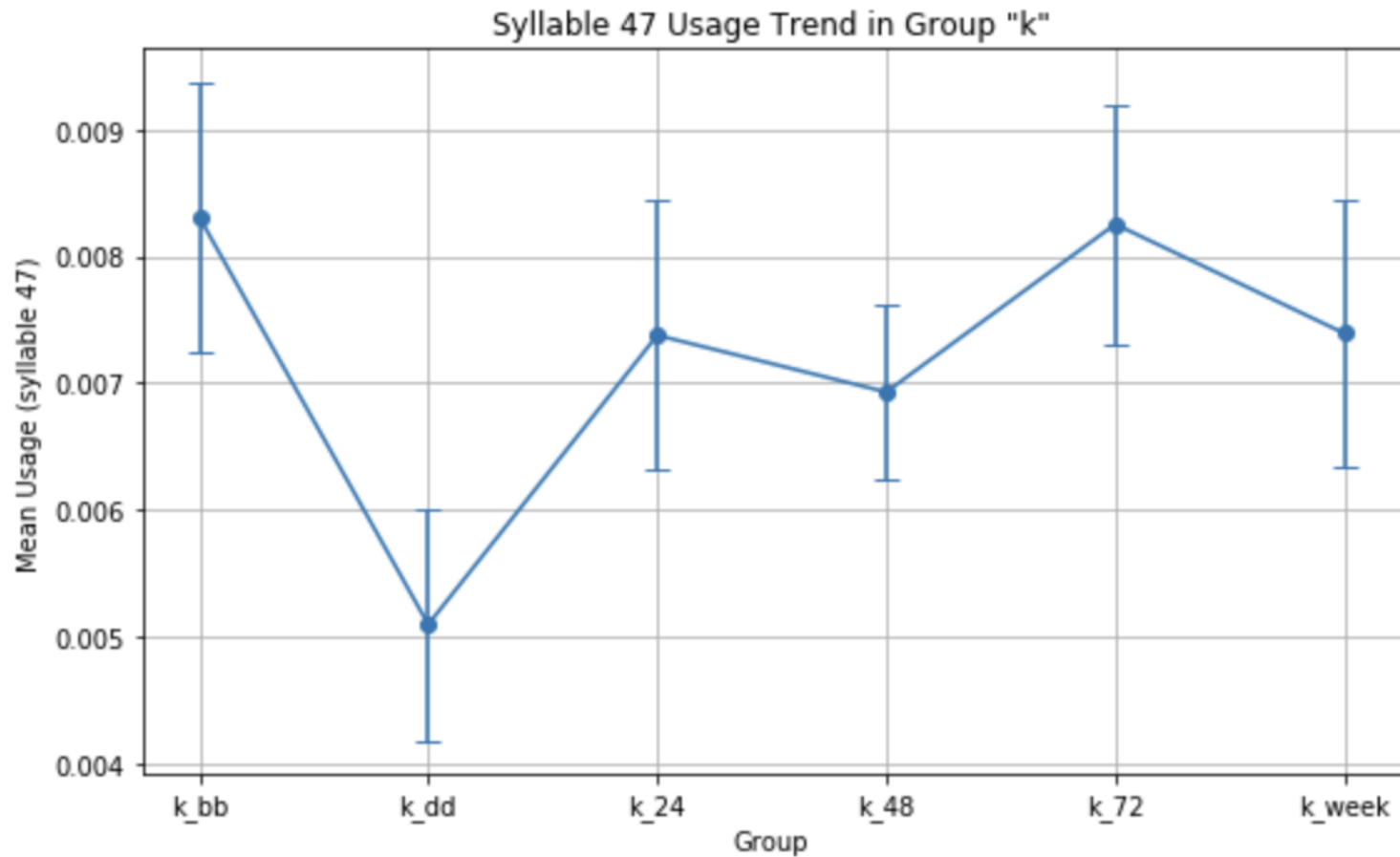
Syllable 32 trend in the ketamine group (Grooming)



Syllable 35 trend in the ketamine group



Syllable 47 trend in the ketamine group (rearing)



Syllable 4 trend in the ketamine group

